

2023 Ch H1 Q20

Section: Nature's Chemistry

Topic: Systematic Carbon Chemistry

Question Summary

Which of the following compounds would react with sodium hydroxide solution to form a salt?

A: CH_3CHO (ethanal)

B: CH_3COOH (ethanoic acid)

C: CH_3COCH_3 (propanone)

D: $\text{CH}_3\text{CH}_2\text{OH}$ (ethanol)

Worked Solution

Sodium hydroxide reacts with acids to form salts and water.

- Aldehydes (CH_3CHO) do not react with NaOH under normal conditions.
- Carboxylic acids (CH_3COOH) react with NaOH to form sodium ethanoate (a salt) and water.
- Ketones (CH_3COCH_3) do not react with NaOH .
- Alcohols ($\text{CH}_3\text{CH}_2\text{OH}$) do not react with

NaOH without strong heating and special conditions.

Final Answer

B — CH₃COOH (ethanoic acid)

Revision Tips

- Only carboxylic acids react readily with alkalis to form salts.
- Reaction: $\text{CH}_3\text{COOH} + \text{NaOH} \rightarrow \text{CH}_3\text{COONa} + \text{H}_2\text{O}$.
- Alcohols and aldehydes do not neutralise bases under normal conditions.